

Double Integrals

Def Let $F(x, y)$ be a continuous function in a closed region R in the xy -plane. Divide R into n subregions ΔA_k of area ΔA_k , $k=1, 2, 3, \dots$

Let also (ξ_k, η_k) be some points of ΔA_k

$$\text{Form the sum } \sum_{k=1}^n F(\xi_k, \eta_k) \Delta A_k \quad \text{--- (i)}$$

$$\text{Consider } \lim_{n \rightarrow \infty} \sum_{k=1}^n F(\xi_k, \eta_k) \Delta A_k \quad \text{--- (ii)}$$

Such that the linear dimensions of ΔA_k approach 0. If this limit exist it is denoted by $\iint_R F(x, y) dA = \iint_R F(x, y) dx dy$ and is called the double integral of $F(x, y)$ over the Reg.

Iterated Integrals

$$\begin{aligned} \iint_R dx dy &= \int_a^c \left[\int_{a_1}^{x_2} dx \right] dy \\ &= \int_a^b \left[\int_{y=y(x)}^{y_2(x)} dy \right] dx \end{aligned}$$

The double integrals are often useful in evaluating Area and Volumes.

Double Integrals

The expression $\int_y^{y_2} \int_{x_1}^{x_2} f(x,y) dx dy$ is called a double integral and indicates

that (i) $f(x,y)$ is first integrated wrt x (regarding y as being constant) between the limit $x=x_1$ and $x=x_2$

(ii) the result is then integrated wrt y between the limits $y=y_1$ and $y=y_2$.

✓ Example: Evaluate $I = \int_1^2 \int_2^4 (x+2y) dx dy$ * Evaluate $\int_R \int (x^2+y^2) dx dy$

$$\Rightarrow I = \int_1^2 \left[\int_2^4 (x+2y) dx \right] dy$$

$$= \int_1^2 \left[\frac{x^2}{2} + 2xy \right]_2^4 dy$$

$$= \int_1^2 [(8+8y) - (2+4y)] dy$$

$$= \int_1^2 (6+4y) dy = [6y+2y^2]_1^2$$

$$= (12+8) - (6+2)$$

$$= 20-8$$

$$= 12$$

where R is the region b/w the line $y=x$ and the curve $y=x^2$

$$(x^2 - x = 0 \Rightarrow x(x-1))$$

$$x(x-1)=0$$

$$x=0, x=1$$

$$\& \text{ if } y=x \Rightarrow y=0$$

$$\frac{-22}{+5}$$

Ex.

Evaluate:

(a) $\int_1^2 \int_3^5 dy dx$

(c) $\int_0^5 \int_1^2 (3x^2-4) dx dy$

(b) $\int_0^4 \int_1^{3x} dy dx$

P.T.O.

$$\begin{aligned}
 \text{(a) } I &= \int_1^2 \int_3^5 dy dx = \int_1^2 \left[y \right]_3^5 dx \\
 &= \int_1^2 (5-3) dx = \int_1^2 2 dx \\
 &= \left[2x \right]_1^2 = 4-2 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } I &= \int_0^4 \int_1^{3x} 2xy dy dx \\
 &= \int_0^4 \left[y^2 \right]_1^{3x} dx \\
 &= \int_0^4 (9x^2 - 1) dx \\
 &= \left[3x^3 - x \right]_0^4 \\
 &= 192 - 4 = 188
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) } I &= \int_0^5 \int_1^2 (3x^2 - 4) dx dy \\
 &= \int_0^5 \left[x^3 - 4x \right]_1^2 dy \\
 &= \int_0^5 (8-8) - (4-4) dy \\
 &= \int_0^5 3 dy = \left[3y \right]_0^5 \\
 &= 15
 \end{aligned}$$

Triple Integrals

Sometimes we have to deal with expressions such as:

$$I = \int_a^b \int_c^d \int_e^f f(x, y, z) dx dy dz$$

but the rules ^{are} as in double integral. Start with inner most integral and work outwards.

$$\int_a^b \left[\int_c^d \left(\int_e^f f(x, y, z) dx \right) dy \right] dz \quad \textcircled{1} \textcircled{2} \textcircled{3}$$

Also ~~the~~ symbols are regarded as constant except the one variable with respect to which stage of integration is taking place.

① Evaluate $I = \int_1^3 \int_1^1 \int_0^2 (x+2y-z) dx dy dz$ ② $I = \int_1^3 \int_0^1 \int_0^3 (x^2+y^2-z^2) dx dy dz$

$$\Rightarrow I = \int_1^3 \int_{-1}^1 \int_0^2 (x+2y-z) dx dy dz$$

$$= \int_1^3 \int_{-1}^1 \left[x^2 + 2xy - xz \right]_0^2 dy dz$$

$$= \int_1^3 \int_{-1}^1 (2+4y-2z) dy dz = \int_1^3 [2y+2y^2-2yz]_{-1}^1 dz$$

$$= \int_1^3 [(2+2-2z) - (-2+2+2z)] dz$$

$$= \int_1^3 (4-4z) dz$$

$$= [4z - 2z^2]_1^3 = (12-18) - (4-2) = -8$$

Ex.

Evaluate

$$\int_1^2 \int_0^3 \int_0^1 (p^2 + q^2 - r^2) dp dq dr$$

Because

$$I = \int_1^2 \int_0^3 \int_0^1 (p^2 + q^2 - r^2) dp dq dr$$

$$= \int_1^2 \int_0^3 \left[\frac{p^3}{3} + pq^2 - pr^2 \right]_0^1 dq dr$$

$$= \int_1^2 \int_0^3 \left\{ \frac{1}{3} + q^2 - r^2 \right\} dq dr$$

$$= \int_1^2 \left[\frac{q}{3} + \frac{q^3}{3} - qr^2 \right]_0^3 dr$$

$$= \int_1^2 (1 + 9 - 3r^2) dr$$

$$= \left[10r - r^3 \right]_1^2 = (20 - 8) - (10 - 1) = 12 - 9 = 3$$

Ex. Evaluate

(a) $I = \int_1^2 \int_3^5 dy dx$

$$= \int_1^2 [y]_3^5 dx$$

$$= \int_1^2 (5 - 3) dx$$

$$= \int_1^2 2 dx$$

$$= \left[2x \right]_1^2 = 4 - 2 = 2$$

b, $\int_0^4 \int_1^{3x} 2y dy dx$

$$I = \int_0^4 \int_1^{3x} 2y dy dx$$

$$= \int_0^4 [y^2]_1^{3x} dx$$

$$= \int_0^4 (9x^2 - 1) dx$$

$$= \left[3x^3 - x \right]_0^4$$

$$= 192 - 4 = 188$$

Scalar and Vector Fields

If corresponding to each value of a scalar u we associate a vector $\vec{A}$, then $\vec{A}$ is called a fn of u denoted by $\vec{A}(u)$. In three dimensions we write $\vec{A}(u) = A_1(u)\vec{i} + A_2(u)\vec{j} + A_3(u)\vec{k}$. Thus, if to each pt (x, y, z) there corresponds a vector $\vec{A}$. Then $\vec{A}$ is a fn of (x, y, z) , indicated by $A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$.

We sometimes say that a vector function $\vec{A}$ defines a vector field since it associates a vector with each pt of a region. Similarly $\phi(x, y, z)$ defines a scalar field since it associates a scalar with each pt of a region.

Definition

Consider the differential operator ∇ or (del) defined by

$$\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}.$$

If $\phi(x, y, z)$ and $\vec{A}(x, y, z)$ have continuous first partial derivatives in a region R then we can define the following.

1. Gradient. The gradient of ϕ is defined by

$$\begin{aligned} \text{grad } \phi &= \nabla \phi = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \phi \\ &= \vec{i} \frac{\partial \phi}{\partial x} + \vec{j} \frac{\partial \phi}{\partial y} + \vec{k} \frac{\partial \phi}{\partial z} \\ &= \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} \end{aligned}$$

2. Divergence. The divergence of $\vec{A}$ is defined by

$$\begin{aligned} \text{div } \vec{A} &= \nabla \cdot \vec{A} = \left(\vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z} \right) \cdot A_1 \vec{i} + A_2 \vec{j} + A_3 \vec{k} \\ &= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \end{aligned}$$

B

3. Curl. The curl of $\vec{A}$ is defined by

$$\begin{aligned}\text{Curl } \vec{A} &= \nabla \times \vec{A} \\ &= \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \times A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k} \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\ &= \hat{i} \left| \frac{\partial}{\partial y} \frac{\partial}{\partial z} \right| - \hat{j} \left| \frac{\partial}{\partial x} \frac{\partial}{\partial z} \right| + \hat{k} \left| \frac{\partial}{\partial x} \frac{\partial}{\partial y} \right| \\ &= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \hat{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \hat{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \hat{k}\end{aligned}$$

4. Directional Derivatives.

The directional derivatives of a function ϕ in direction of $\underline{n}$ is defined as $\nabla \phi \cdot \underline{n}$

C

4. Directional Derivatives

The directional derivative of a fn ϕ in the direction of $\underline{n}$ is defined as $\nabla\phi \cdot \underline{n}$

Formulas involving ∇

If the partial derivatives of $\vec{A}$, $\vec{B}$, u , and v are assumed to exist, then

1. $\text{grad}(u+v) = \text{grad } u + \text{grad } v$ or $\nabla(u+v) = \nabla u + \nabla v$
2. $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$ or $\text{div}(\vec{A} + \vec{B}) = \text{div } \vec{A} + \text{div } \vec{B}$
3. $\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$ or $\text{Curl}(\vec{A} + \vec{B}) = \text{Curl } \vec{A} + \text{Curl } \vec{B}$
4. $\nabla(u\vec{A}) = (\nabla u) \cdot \vec{A} + u(\nabla \cdot \vec{A})$
5. $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$
6. $\nabla \cdot (\nabla u) = \nabla^2 u$

$\equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$ is called the Laplacian of u .

And $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called the Laplacian operator

~~Q~~ Grad, div, & curl.

If $\phi = x^2yz^3$ and $\vec{A} = xz\vec{i} + y^3\vec{j} + 2x^2y\vec{k}$, find (i) $\nabla\phi$, (ii) $\nabla \times \vec{A}$,
(iv) $\text{div}(\phi\vec{A})$ (v) $\text{curl}(\phi\vec{A})$

a, $\nabla\phi = \left(\frac{\partial}{\partial x}\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\frac{\partial}{\partial y} + \frac{\partial}{\partial z}\frac{\partial}{\partial z}\right)\phi$

$$= \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k}$$

$$= \frac{\partial}{\partial x}(x^2yz^3)\vec{i} + \frac{\partial}{\partial y}(x^2yz^3)\vec{j} + \frac{\partial}{\partial z}(x^2yz^3)\vec{k}$$

$$= 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$$

b, $\nabla \cdot \vec{A} = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \cdot (xz\vec{i} - y^2\vec{j} + 2x^2y\vec{k})$

$$= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(2x^2y)$$

$$= z - 2y$$

c, $\nabla \times \vec{A} = \left(\frac{\partial}{\partial x}\frac{\partial}{\partial x} + \frac{\partial}{\partial y}\frac{\partial}{\partial y} + \frac{\partial}{\partial z}\frac{\partial}{\partial z}\right) \times (xz\vec{i} - y^2\vec{j} + 2x^2y\vec{k})$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & -y^2 & 2x^2y \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y}(2x^2y) - \frac{\partial}{\partial z}(-y^2)\right)\vec{i} + \left(\frac{\partial}{\partial z}(xz) - \frac{\partial}{\partial x}(2x^2y)\right)\vec{j}$$

$$+ \left(\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial y}(xz)\right)\vec{k}$$

$$= 2x^2\vec{i} + (x - 4xy)\vec{j}$$

$$d. \operatorname{div}(\phi_A) = \nabla \cdot (\vec{\phi}_A)$$

$$\begin{aligned}
 &= \nabla \cdot (x^3 y z^4 \vec{i} - x^2 y^3 z^3 \vec{j} + 2x^4 y^2 z^3 \vec{k}) \\
 &= \frac{\partial}{\partial x} (x^3 y z^4) + \frac{\partial}{\partial y} (-x^2 y^3 z^3) + \frac{\partial}{\partial z} (2x^4 y^2 z^3) \\
 &= 3x^2 y z^4 - 3x^2 y^2 z^3 + 6x^4 y^2 z^2
 \end{aligned}$$

$$e. \operatorname{Curl}(\vec{\phi}_A) = \nabla \times (\vec{\phi}_A)$$

$$= \nabla \times (x^3 y z^4 \vec{i} - x^2 y^3 z^3 \vec{j} + 2x^4 y^2 z^3 \vec{k})$$

$$= \begin{vmatrix} \vec{i} & \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^3 y z^4 & -x^2 y^3 z^3 & 2x^4 y^2 z^3 \end{vmatrix}$$

$$= (4x^4 y z^3 - 3x^2 y^3 z^2) \vec{i} + (4x^3 y z^3 - 8x^3 y^2 z^3) \vec{j} - (2x y^3 z^3 + 2x^2 y^2 z^3) \vec{k}$$

Scalar and Vector fields

Ex. 1. If $\phi = 2x^2y - xz^3$, find (i) $\nabla\phi$ (ii) $\nabla^2\phi$

2. If $\phi = xy + yz + zx$ and $\vec{A} = x^2y\vec{i} + y^2z\vec{j} + z^2x\vec{k}$

Find (i) $\vec{\nabla}\phi$, (ii) $\phi \vec{\nabla} \cdot \vec{A}$ and (iii) $(\vec{\nabla}\phi) \times \vec{A}$ at the

$(3, -1, 2)$.

$56\vec{i} - 30\vec{j} + 47\vec{k}$

Vector Notation

It is sometimes convenient to express a line line integral in Vector form as aid in physical or geometrical understanding as well as for brevity of notation

$$\text{i.e. } \int_C P dx + Q dy + R dz = \int_C [P(x, y, z)\vec{i} + Q(x, y, z)\vec{j} + R(x, y, z)\vec{k}] \cdot (dx\vec{i} + dy\vec{j} + dz\vec{k})$$

$$= \int_C \vec{S} \cdot d\vec{r}$$

$$\text{Where } \vec{S} = P\vec{i} + Q\vec{j} + R\vec{k}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

If we associate a force on a particle at each point of C , the line integral physically becomes the total work done in moving the particle along C .

$$\text{i.e. } W = \int_C F \cdot d\vec{r}$$

If C is closed then we write $\oint$ instead of $\int$

If the $\vec{S} = 0$ i.e. $\nabla \times \vec{S} = 0$ then (i) becomes equal to zero

$$\text{i.e. } \oint_C \vec{S} \cdot d\vec{r} = 0 \quad \text{--- (ii)}$$

In rectangular form we can write (ii) as $\oint P dx + Q dy = 0$

$$\text{If } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

or if the integrand is exact.

Eg. Evaluate the line integral

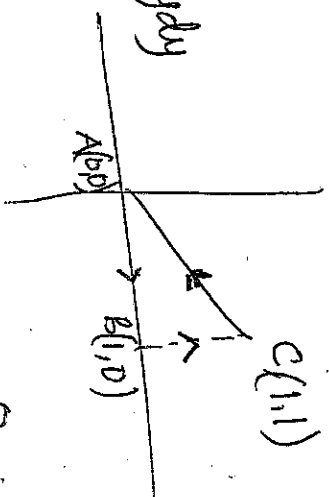
$$\int_C (x^2 + y^2) dx + 2xy dy \quad \text{(i) If } C \text{ is the triangle joining the pts } (0,0), (1,0) \text{ and } (1,1)$$

? (ii) If C : the parabola $x=t, y=t^2, t=0, t=1$

B

$$(i) \oint_C (x^2+y^2)dx + 2xydy$$

$$= \int_{AB} (x^2+y^2)dx + 2xydy + \int_{BC} (x^2+y^2)dx + 2xydy + \int_{CA} (x^2+y^2)dx + 2xydy$$



$$AB: y=0$$

$$BC: x=1$$

$$dx=0$$

$$CA: y=x$$

$$= \int_0^1 x^2 dx + \int_0^1 2xy dy + \int_1^0 4y^2 dy$$

$$= \left. \frac{x^3}{3} \right|_0^1 + \left. y^2 \right|_0^1 + \left. \frac{4y^3}{3} \right|_1^0$$

$$= \frac{1}{3} + 1 - \frac{4}{3}$$

$$= \frac{1}{3} - \frac{4}{3} = 0$$

$$(ii) \int_0^1 \left(\int_0^1 (t^3 + (t^2+1)^2) dx + 2t(t^2+1) 2t dt \right)$$

$$= \int_0^1 (t^3 + t^4 + 2t^2 + 1) 2t dt + 4t^4 + 4t^2 dt$$

$$= \int_0^1 (2t^5 + 4t^3 + 2t^2 + 2t - 1) dt$$

=

Maths

Green's Theorem

Let $P(x,y)$, $Q(x,y)$, $\frac{\partial P}{\partial y}$, $\frac{\partial Q}{\partial x}$ be single valued and continuous in a simply connected region R bounded by a simple closed curve C . Then

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

Ex. Verify the Green's Theorem in the plane for $\oint_C (2xy - x^2)dx + (x + y^2)dy$ where

C is the closed curve bounding the region b/w $y=x^2$ and $x=y^2$. Where $\oint$ is used to emphasize that C is closed and that it is described in the positive direction.

$$\oint_C (2xy - x^2)dx + (x + y^2)dy = \int_0^1 (2x^3 - x^2)dx + \int_1^0 (x + x^4)2xdx$$

Along $y=x^2$ & $y^2=x$ intersect at $(0,0)$ & $(1,1)$

Along $y=x^2$ the line integral $\int_0^1 (2x^3 + x^2 + 2x^5)dx + \int_1^0 (4y^4 - 2xy^5 + 2xy^2)dy$

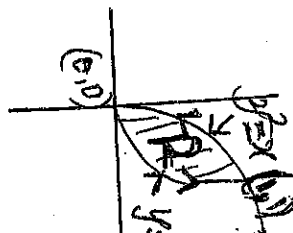
$$= \int_0^1 (2x^3 + x^2 + 2x^5)dx + \int_1^0 (4y^4 - 2xy^5 + 2xy^2)dy$$

$$= \left(\frac{2x^4}{4} + \frac{x^3}{3} + \frac{2x^6}{6} \right) \Big|_0^1 + \left(\frac{4y^5}{5} - \frac{xy^6}{3} + \frac{2xy^3}{3} \right) \Big|_1^0$$

$$= \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{3} \right) - \left(\frac{4}{5} - \frac{1}{3} + \frac{2}{3} \right)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{4}{5} = \frac{15+10-24}{30}$$

$$\text{i.e. } \frac{25-24}{30} = \frac{1}{30}$$



Now 1st $\int_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \quad \Big|_{R=dy}^{P=dx}$

$$= \int \int (1-2x) dx dy$$

$$= \int_0^1 \left[\int_{x=y^2}^{x=y} (1-2x) dx \right] dy$$

$$= \int_0^1 (x-x^2) \Big|_{y^2}^{y} dy$$

$$= \int_0^1 (y^2-y) - (y^2-y^4) dy$$

$$= \left[\frac{2y^3}{3} - \frac{y^2}{2} - \frac{y^3}{3} + \frac{y^5}{5} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{2} - \frac{1}{3} + \frac{1}{5}$$

$$= \frac{1}{3} + \frac{1}{5} - \frac{1}{2}$$

$$= \frac{16-15}{30} \quad \text{or} \quad \frac{16}{30} - \frac{15}{30} = \frac{1}{30}$$

$\therefore$ Green's Theorem Verified since LHS = $\frac{1}{30}$, RHS = $\frac{1}{30}$

The Plane Curves $y=x^2$ and $y^2=x$ intersect at $(0,0)$ and $(1,1)$

Along $y=x^2$ the line integral equals

$$\int_{x=0}^1 (2x(x^2) - x^2) dx + \int_0^1 (x + (x^2)^2) dx - 2x dx$$

$$= \int_0^1 (2x^3 + x^2 - 2x^5) dx + \text{Along } y^2=x + \int_1^0 (2xy^2xy) - (y^2)^2 dy + (y^2 + y^2) dy$$

$$+ \int_1^0 (4y^4 - 2y^5 + 2y^2) dy$$

$$\therefore \int_0^1 (2x^3 + x^2 - 2x^5) dx + \int_1^0 (4y^4 - 2y^5 + 2y^2) dy$$

$$= \left(\frac{2x^4}{4} + \frac{x^3}{3} - \frac{2x^6}{6} \right) \Big|_0^1 + \left(\frac{4y^5}{5} - \frac{y^6}{3} + \frac{2y^3}{3} \right) \Big|_1^0$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{3} - \left(\frac{4}{5} - \frac{1}{3} + \frac{2}{3} \right)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{4}{5} = \frac{15 + 10 - 24}{30}$$

$$= \frac{25 - 24}{30}$$

$$= \frac{1}{30}$$

The Divergence Theorem

Q. Let A be a Vector field that is continuously differentiable on closed-spaced region V bounded by a smooth Surface S . Then

$$\iiint_V \nabla \cdot \vec{A} dV = \iint_S A \cdot \vec{n} ds \quad \text{--- (i) where } \vec{n} \text{ is outwardly drawn}$$

Normal.

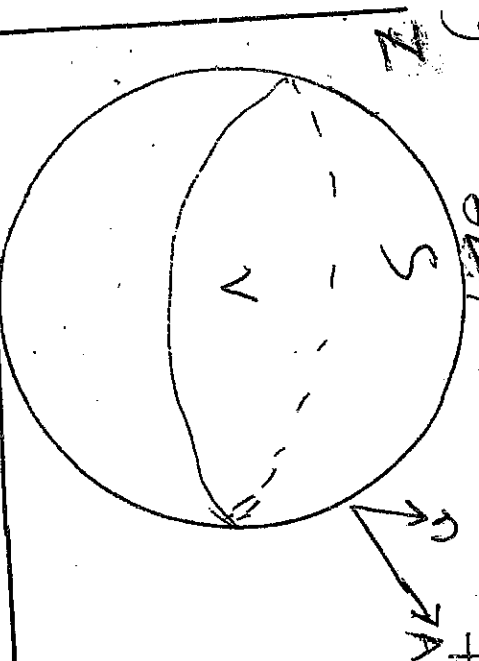
If $\vec{n}$ is expressed through direction cosines i.e.

$$\vec{n} = \hat{i} \cos \alpha + \hat{j} \cos \beta + \hat{k} \cos \gamma, \text{ then equation (i) may be}$$

Written as:

$$\iiint_V \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) dV = \iint_S (A_1 \cos \alpha + A_2 \cos \beta + A_3 \cos \gamma) dS$$

The Cartesian Component form of Equation (2) is

$$\iiint_V \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) dV = \iint_S (A_1 dy dz + A_2 dz dx + A_3 dx dy)$$


The Divergence theorem establishes equality b/w a triple integral (Volume integral) of a function over a region of three-dimensional space and a double integral of the function over the surface S.

The Surface that bounds that region. This relation is
Very important in the expression of physical theories